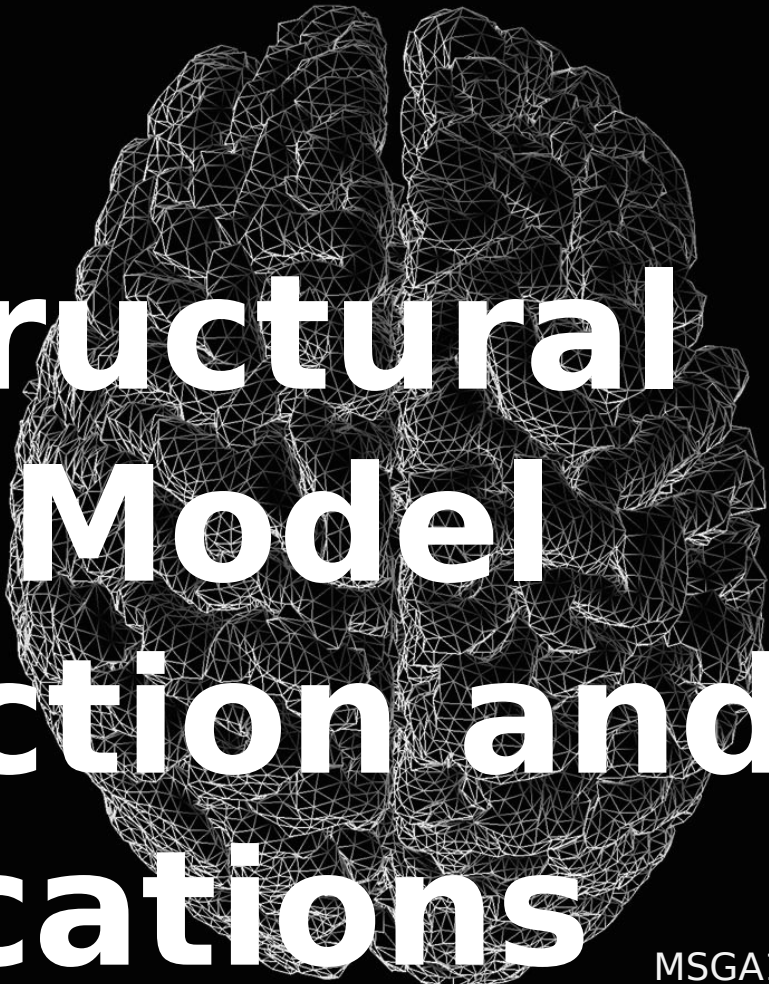




3D Structural Brain Model Extraction and Applications

MSG18 23 - 02 Keiland W. Cooper

LetterWriting_122B
ID: 8007@khjames/LetterWriting
* 1/1/1996
Study 1
11/5/2015
3:24:13 PM
79 IMA

Indiana University Psychology Department
TrioTim
HFS

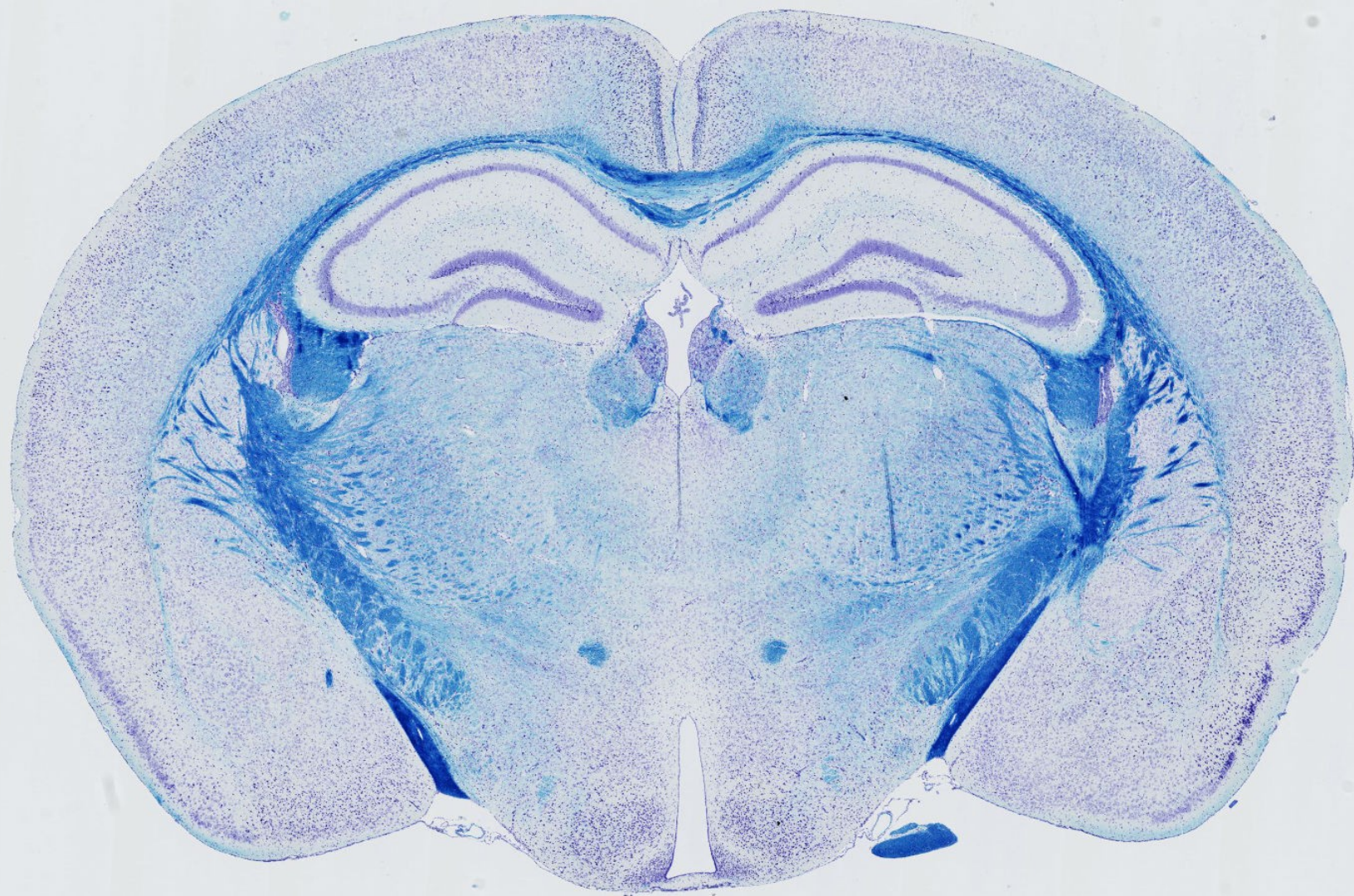


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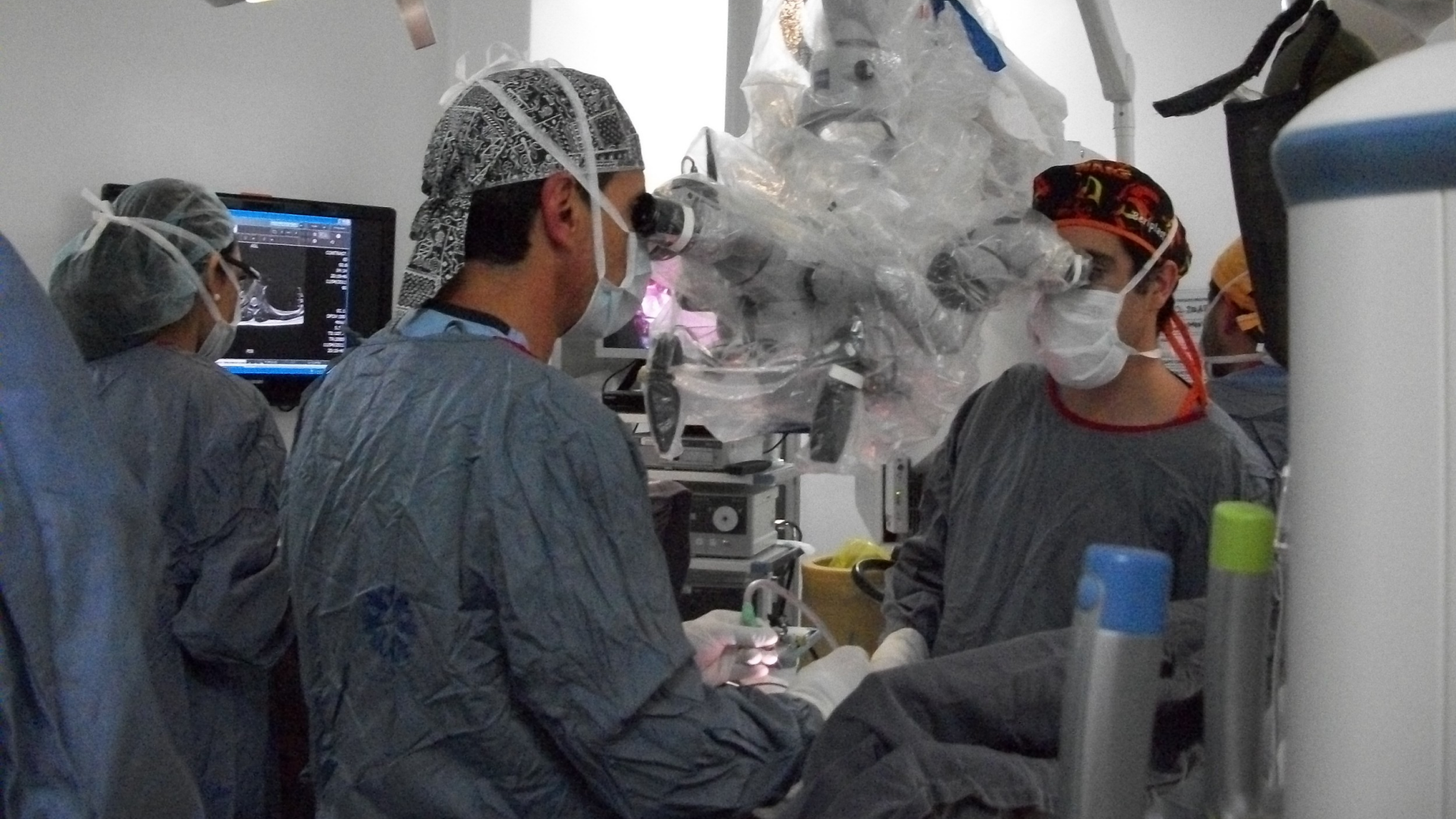
SL 1
TE 2.67
TR 1800
TI 900

SP R1.5
FoV 256*256
256*256
Sag
W: 712
C: 319

**The brain
is not flat.**









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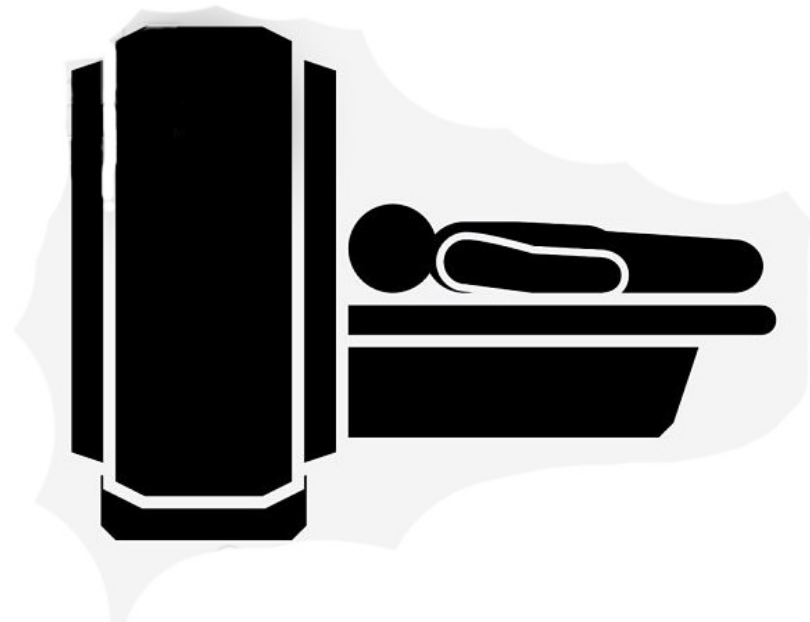
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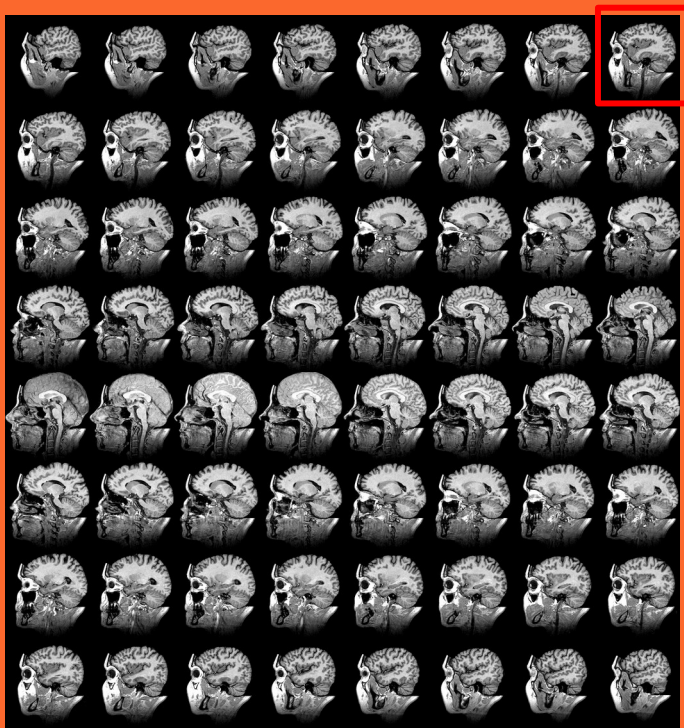
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SP R1.5
FoV 256*256
256*256
Sag
W: 712
C: 319

fMRI:

a crash course



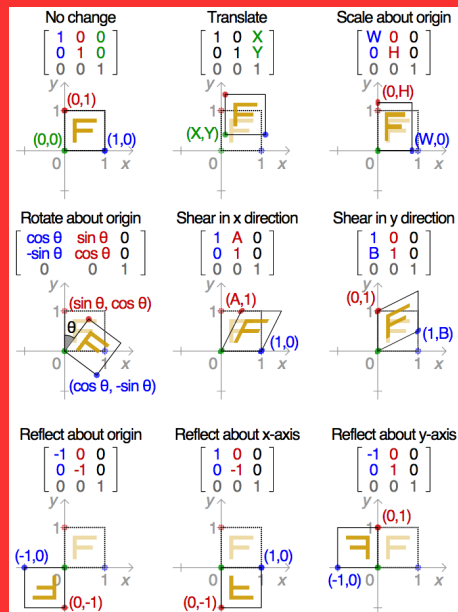


$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{bmatrix}_{n \times m}$$

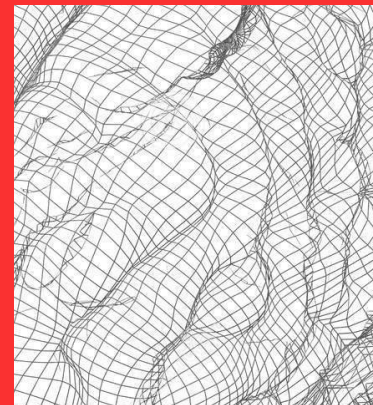
Pipeline



Brain Localization /
Skull stripping



Transformation /
Parameter exploration



Mesh Construction

Pipeline

That is, the prior probability of a label at a given vertex $\mathbf{r}$ is only influenced by the labels within some neighborhood of $\mathbf{r}$. The locality restriction imposed by the Markov model permits the probability of the entire parcellation to be written in terms of neighborhood or *clique* potentials $V_c(P)$ via the Hammersley–Clifford theorem (Besag, 1974). That is, the probability $p(P)$ can be equivalently characterized by a Gibbs distribution:

$$p(P) = \frac{1}{Z} e^{-U(P)} \quad (11)$$

where Z is a normalizing constant and will be dropped in the following, and $U(P)$ is an energy function that can be written in the form:

$$U(P) = \sum_c V_c(P) \quad (12)$$

The clique potentials $V_c(P)$ encode the energy associated with a certain configuration of labels within the c th clique. Choosing $V_c(P)$ to be $-\log(p(P(\mathbf{r})|P(\mathbf{r}_1), P(\mathbf{r}_2) \dots P(\mathbf{r}_K)))$, where $\mathbf{r}$ is the central vertex of the c th clique, allows us write the probability of the entire parcellation as the product of the probability of the label at each vertex, given its neighborhood:

$$p(P) = \prod_{\mathbf{r} \in S} p(P(\mathbf{r})|P(\mathbf{r}_1), P(\mathbf{r}_2), \dots, P(\mathbf{r}_K)), \mathbf{r}_i \in N(\mathbf{r}) \quad (13)$$

Using Bayes rule, we can rewrite this as:

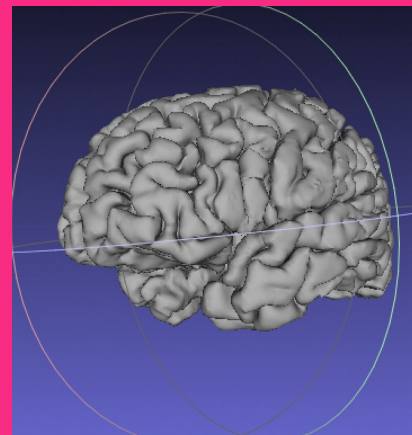
$$p(P) \propto \prod_{\mathbf{r} \in S} p(P(\mathbf{r})) p(P(\mathbf{r}_1), P(\mathbf{r}_2), \dots, P(\mathbf{r}_K)|P(\mathbf{r})), \mathbf{r}_i \in N(\mathbf{r}) \quad (14)$$

Equation (14) allows the probability of a given label to be modulated by any configuration of neighboring labels. While this would be extremely useful, it is unfortunately not computationally tractable to implement, as one would need to compute separate prior probabilities for every combination of neighboring labels that occur. Instead, we make the simplifying assumption that

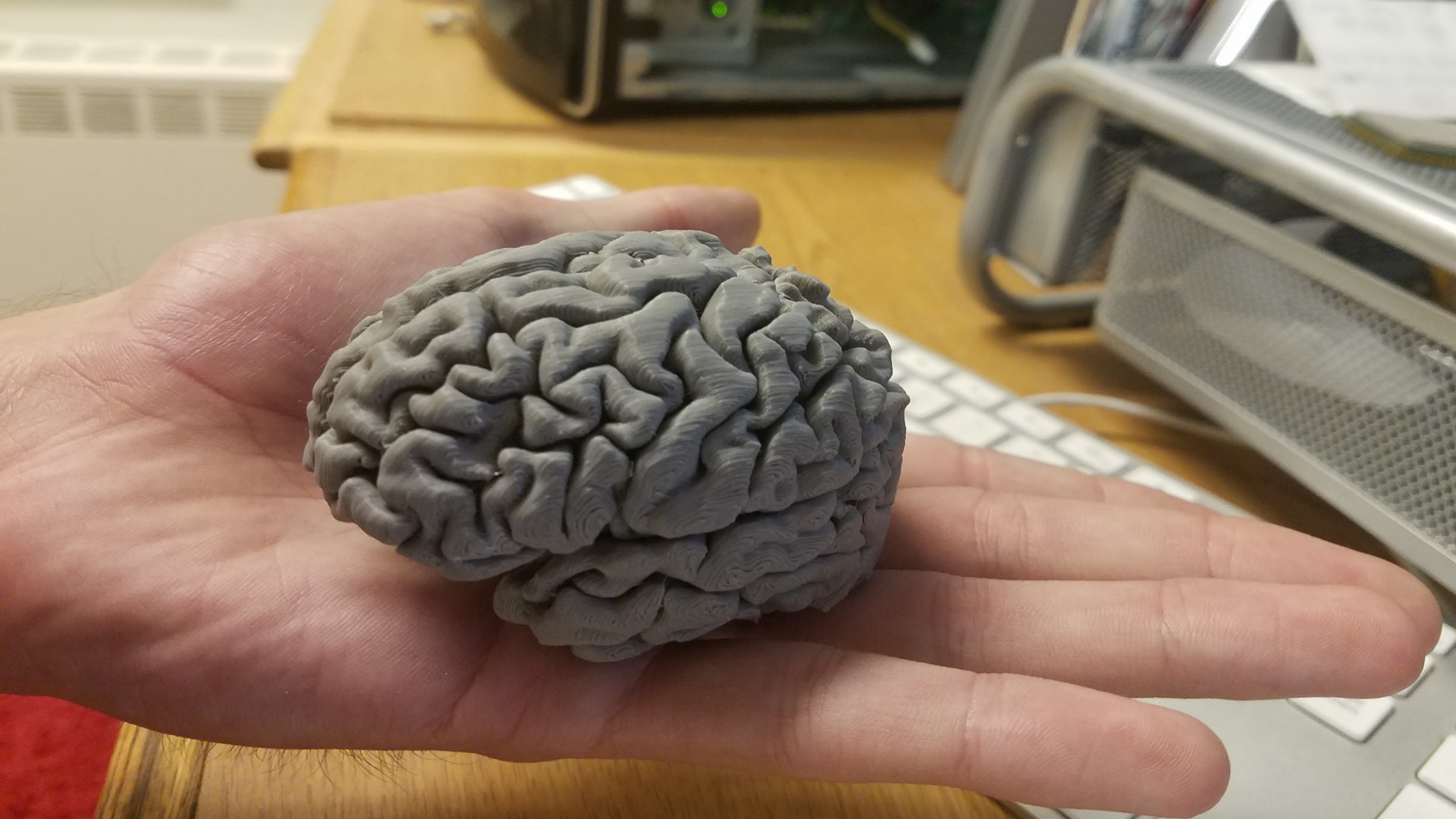
Pipeline

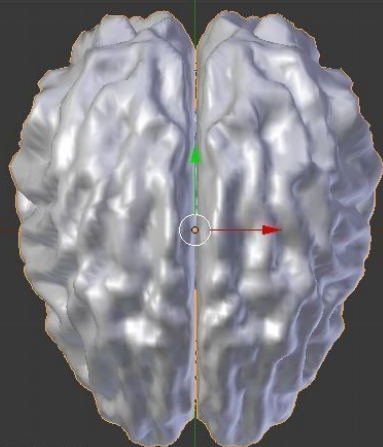


Native File Conversion

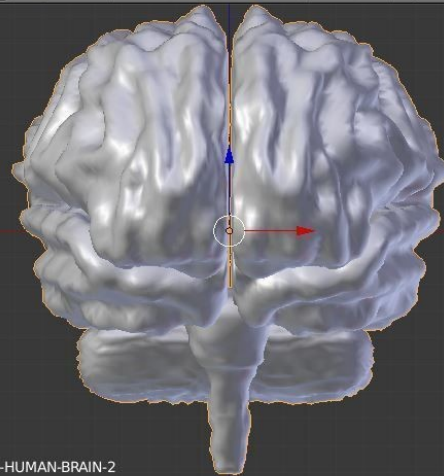


Post processing / rendering

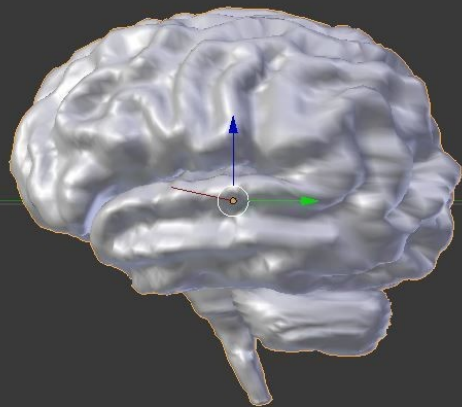


Top Ortho
Centimeters

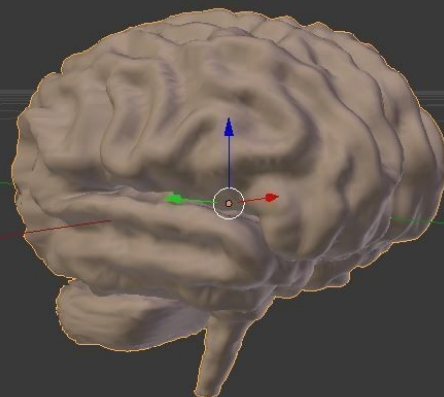
(0) ANATOMY---HUMAN-BRAIN-2

Front Ortho
Centimeters

(0) ANATOMY---HUMAN-BRAIN-2

Right Persp
Meters

(0) ANATOMY---HUMAN-BRAIN-2

User Persp
Meters

(0) ANATOMY---HUMAN-BRAIN-2

View Search All Scenes

Scene

RenderLayers

World

ANATOMY---HUMAN-BRAIN

ANATOMY---HUMAN-BRAIN-2

ANATOMY---HUMAN-BRAIN-2

ANATOMY---HUMAN-BRAIN-2

ANATOMY---HUMAN-BRAIN-2

ANATOMY---HUMAN-BRAIN-2

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ANATOMY---HUMAN-BRAIN-2

ANATOMY---HUMAN-BRAIN-2





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Ref.

Thank You

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